

Probability

Few centuries ago, gambling and gaming were considered to be fashionable and became widely popular among many men. As the games became more complicated, players were interested in knowing the chances of winning or losing a game from a given situation. In 1654, Chevalier de Mere, a French nobleman with a taste of gambling, wrote a letter to one of the prominent mathematician of the time, Blaise Pascal, seeking his advice about how much dividend he would get for a gambling game played by paying money. Pascal worked this problem mathematically but thought of sharing this problem and see how his good friend and mathematician Pierre de Fermat could solve. Their subsequent correspondences on the issue represented the birth of Probability Theory as a new branch of mathematics.



Basic concepts of Probability

i) Random Experiment

If an experiment or trial can be repeated under the same conditions, any number of times and it is possible to count the total number of outcomes, but individual result ie., individual outcome is not predictable, then the experiment is known as random experiment.

Example: Tossing a coin, throwing a die, selecting a card from a pack of playing cards, etc.

(ii) Outcome:

The result of a random experiment will be called an outcome, events .

(iii) Trial and Event:

Any particular performance of a random experiment is called a trial and outcome or combinations of outcomes are termed as events.

(iv) Exhaustive Events:

The total number of possible outcomes of a random experiment is known as the exhaustive events.

(v) Favourable Events:

The number of cases favourable to an event in a trial is the number of outcomes which entail the happening of the event.

(vi) Mutually Exclusive events:

Events are said to be mutually exclusive if the happening of any one of them precludes the happening of all the others, ie., if no two or more of them can happen simultaneously in the same trial. Symbolically the event A and B are mutually exclusive if $A \cap B = \emptyset$.

(vii) Equally Likely Events:

Events (two or more) of an experiment are said to be equally likely, if any one of them cannot be expected to occur in preference to the others.

Biased unbiased

(viii) Classical definition of Probability

If a random experiment or trial results in ‘ n ’ exhaustive, mutually exclusive and equally likely outcomes (or cases), out of which m are favourable to the occurrence of an event E , then the probability ‘ p ’ of occurrence (or happening) of E , usually denoted by $P(E)$, is given by

$$P = P(E) = \frac{\text{Number of favourable cases}}{\text{Total number of exhaustive cases}} = \frac{m}{n}$$

Properties

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(i) $0 \leq P(E) \leq 1$

(ii) Sum of all the probability equal to 1.

(iii) If $P(E)=0$ then E is an impossible event.

For example : A coin is tossed. Find the probability of getting a head

Solution: The total possible outcomes of an experiment {H, T}

Therefore $n = 2$

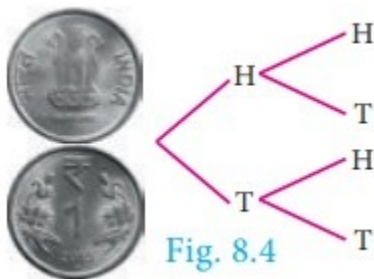
The favorable outcome for getting a head {H}. Therefore $m = 1$. Thus the required probability is

$$P(\text{getting a } H) = \frac{m}{n} = \frac{1}{2}$$

Example : If two coin tossed at a time. Find the probability of getting

i) one head ii)two heads iii)three heads iv)at least two heads v)more than one heads.

When we toss two coins, then from the tree diagram (Fig.8.4), the sample space can be written as $S=\{HH,HT,TH,TT\}$



F(x)= x= 0,1,2

Possible outcomes {HH,HT,TH, TT}

outcomes	No. of Heads
HH	2
HT	1
TH	1
TT	0

Therefore $n=4$ $f(x=1)=1/4$

$P(x=0)$ $(x=0)$ $p(x>=2)=$

$P(\text{one head})= 2/4 =1/2$ ii) $P(\text{two heads})= 1/4$ $p(x=2)=1/4$

iii) $P(\text{three heads}) = 0$ iv) $P(\text{at least two heads}) =1/4$

v) $P(\text{more than one}) =1/4$ vi) $P(\text{less than two heads})=3/4$

Example: The king and queen of diamonds, queen and jack of hearts, jack and king of spades are removed from a deck of 52 playing cards and then well shuffled. Now one card is drawn at random from the remaining cards. Determine the probability that the card is

- (i) a clavor
- (ii) a queen of red card
- (iii) a king of black card

Solution :

In a deck of 52 cards,
the king and queen of diamonds

queen and jack of hearts

jack and king of spades are removed

$$\therefore n(S) = 52 - 6 = 46$$

(i) Let A be the event of getting 'a clavor'

$$\therefore n(A) = 13$$

$$p(A) = \frac{n(A)}{n(S)} = \frac{13}{46}$$

(ii) Let B be the event of getting 'a queen of red card'

$$\therefore n(B) = 0$$

$$p(B) = \frac{n(B)}{n(S)} = \frac{0}{46} = 0$$

(iii) Let C be the event of getting 'a king of black card'

$$\therefore n(C) = 1 \quad [1 \text{ clavor black card}]$$

$$p(C) = \frac{n(C)}{n(S)} = \frac{1}{46}$$

Example: A bag contains 5 red balls, 6 white balls, 7 green balls, 8 black balls. One ball is drawn at random from the bag. Find the probability that the ball drawn is

(i) White

(ii) black or red

(iii) not white

(iv) neither white nor black

Solution :

A bag contains 5 red balls, 6 white balls, 7 green balls, and 8 black balls.

$$\therefore n(S) = 5 + 6 + 7 + 8 = 26$$

(i) Let A be the event of getting 'white balls'

$$n(A) = 6$$

$$p(A) = \frac{n(A)}{n(S)} = \frac{6}{26} = \frac{3}{13}$$

(ii) Let B be the event of getting 'black or red'

$$n(B) = 8 + 5 = 13$$

$$p(B) = \frac{n(B)}{n(S)} = \frac{13}{26} = \frac{1}{2}$$

(iii) Let C be the event of getting 'not a white ball'

$$n(C) = 26 - 6 = 20$$

$$p(C) = \frac{n(C)}{n(S)} = \frac{20}{26} = \frac{10}{13}$$

(iv) Let D be the event of getting 'neither white nor black'

$$n(D) = 5 + 7 = 12$$

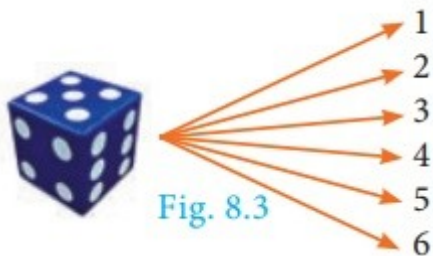
$$p(D) = \frac{n(D)}{n(S)} = \frac{12}{26} = \frac{6}{13}$$

ix) Tree diagram

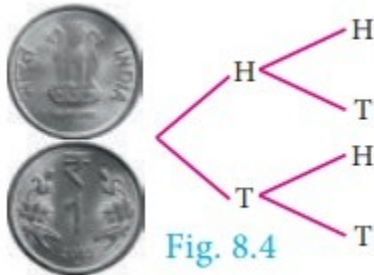
Tree diagram allow us to see visually all possible outcomes of an random experiment. Each branch in a tree diagram represent a possible outcome.

Illustration

(i) When we throw a die, then from the tree diagram (Fig.8.3), the sample space can be written as $S = \{1,2,3,4,5,6\}$

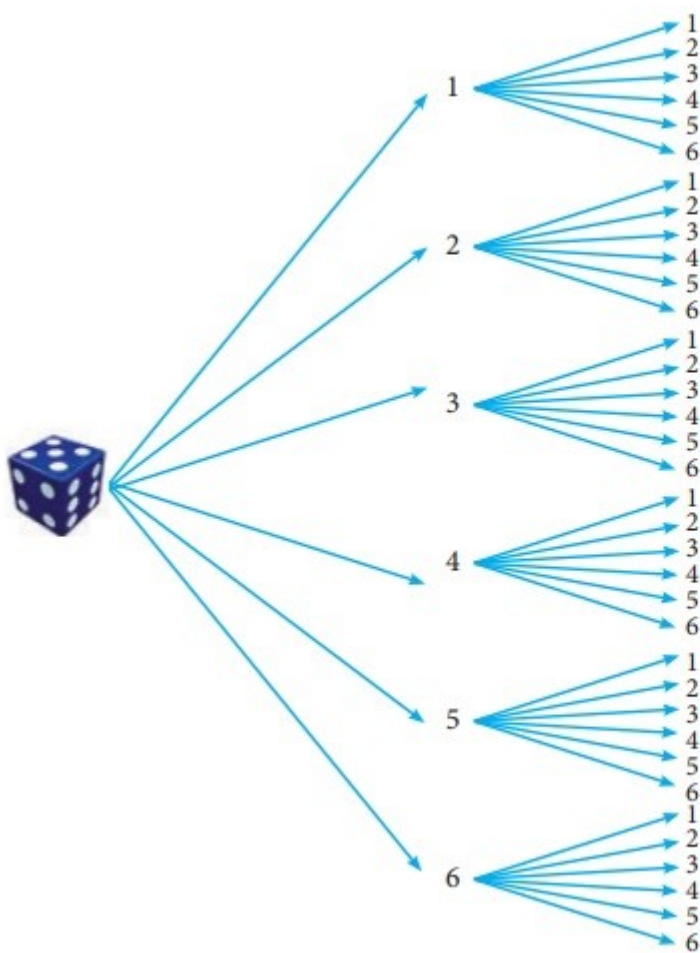


(ii) When we toss two coins, then from the tree diagram (Fig.8.4), the sample space can be written as $S = \{HH, HT, TH, TT\}$



Example Express the sample space for rolling two dice using tree diagram.

Solution When we roll two dice, since each die contain 6 faces marked with 1,2,3,4,5,6 the tree diagram will look like



Hence, the sample space can be written as

$$S = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6)$$

$$(2,1), (2,2), (2,3), (2,4), (2,5), (2,6)$$

$$(3,1), (3,2), (3,3), (3,4), (3,5), (3,6)$$

$$(4,1), (4,2), (4,3), (4,4), (4,5), (4,6)$$

$$(5,1), (5,2), (5,3), (5,4), (5,5), (5,6)$$

$$(6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\}$$

Hence, the sample space can be written as

$$S = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6)$$

$$(2,1), (2,2), (2,3), (2,4), (2,5), (2,6)$$

$$(3,1), (3,2), (3,3), (3,4), (3,5), (3,6)$$

$$(4,1), (4,2), (4,3), (4,4), (4,5), (4,6)$$

$$(5,1), (5,2), (5,3), (5,4), (5,5), (5,6)$$

$$(6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\}$$

(x) Modern Definition of Probability

The modern approach to probability is purely axiomatic and it is based on the set theory concepts. In order to study, the theory of probability with an axiomatic approach it is necessary to define certain basic concepts. They are

(i) Sample Space: Each possible outcome of an experiment that can be repeated under similar or identical conditions is called a sample point and the collection of sample points is called the sample space, denoted by S .

(ii) Event: Any subset of a sample space is called an event.

(iii) Mutually Exclusive events: Two events A and B are said to be mutually exclusive events if $A \cap B = \phi$ i.e., if A and B are disjoint sets.

Example: Consider $S = \{1, 2, 3, 4, 5\}$

Let $A =$ the set of odd numbers $= \{1, 3, 5\}$

and $B =$ the set of even numbers $= \{2, 4\}$

Then $A \cap B = \phi$

Therefore the events A and B are mutually exclusive

(xi) Observation:

Statement meaning in terms of Set theory approach

(i) $A \cup B \Rightarrow$ at least one of the events A or B occurs

(ii) $A \cap B \Rightarrow$ both events A and B occurs

(iii) $\overline{A} \cap \overline{B} \Rightarrow$ Neither A nor B occur

(iv) $A \cap \overline{B} \Rightarrow$ Event A occurs and B does not occur

(xii) Basic Theorems on probability

Theorem 1:

Then $P(\emptyset) = 0$ i.e., probability of an impossible event is zero.

Theorem 2:

Let S be the sample space and A be an event in S then $P(\bar{A}) \equiv 1 - P(A)$

Theorem 3: Addition Theorem (additive law)

i) If the two events A and B are mutually exclusive then $A \cap B = \emptyset$

$$\therefore P(A \cap B) = 0$$

$$\Rightarrow P(A \cup B) = P(A) + P(B) \text{ or } P(A+B) = P(A) + P(B)$$

ii) If A and B are any two events are not mutually exclusive events then (additive law)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

The addition theorem may be extended to any three events A, B, C and we have

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

Proof

(i) Let A and B be any two events of a random experiment with sample space S .

From the Venn diagram, we have the events only A , $A \cap B$ and only B are mutually exclusive and their union is $A \cup B$

Therefore, $P(A \cup B) = P[(\text{only } A) \cup (A \cap B) \cup (\text{only } B)]$

$$= P(\text{only } A) + P(A \cap B) + P(\text{only } B)$$

$$= [P(A) - P(A \cap B)] + P(A \cap B) + [P(B) - P(A \cap B)]$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

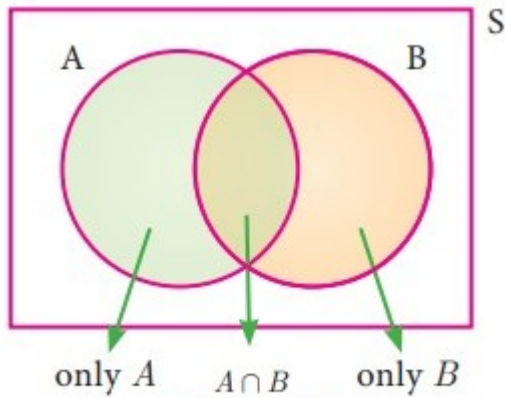


Fig. 8.10

(ii) Let A, B, C are any three events of a random experiment with sample space S .

Let $D = B \cup C$

$$P(A \cup B \cup C) = P(A \cup D)$$

$$= P(A) + P(D) - P(A \cap D)$$

$$= P(A) + P(B \cup C) - P[A \cap (B \cup C)]$$

$$= P(A) + P(B) + P(C) - P(B \cap C) - P[(A \cap B) \cup (A \cap C)]$$

$$= P(A) + P(B) + P(C) - P(B \cap C) - P(A \cap B) - P(A \cap C) + P[(A \cap B) \cap (A \cap C)]$$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

Example

If $P(A) = 0.37$, $P(B) = 0.42$, $P(A \cap B) = 0.09$ then find $P(A \cup B)$.

Solution

$$P(A) = 0.37, P(B) = 0.42, P(A \cap B) = 0.09$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = 0.37 + 0.42 - 0.09 = 0.7$$

Example

What is the probability of drawing either a king or a queen in a single draw from a well shuffled pack of 52 cards?

Solution

Total number of cards = 52

Number of king cards = 4

Probability of drawing a king card = $4/52$ A

Number of queen cards = 4

Probability of drawing a queen card = $4/52$

Both the events of drawing a king and a queen are mutually exclusive

$$\Rightarrow P(A \cup B) = P(A) + P(B)$$

Therefore, probability of drawing either a king or a queen = $4/52 + 4/52 = 2/13$

Example

Two dice are rolled together. Find the probability of getting a doublet or sum of faces as 4.

Solution

When two dice are rolled together, there will be $6 \times 6 = 36$ outcomes. Let S be the sample space. Then $n(S) = 36$

Let A be the event of getting a doublet and B be the event of getting face sum 4.

Then $A = \{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)\}$

$B = \{(1,3), (2,2), (3,1)\}$

Therefore, $A \cap B = \{(2,2)\}$

Then, $n(A) = 6$, $n(B) = 3$, $n(A \cap B) = 1$.

$$P(A) = \frac{n(A)}{n(S)} = \frac{6}{36}$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{3}{36}$$

$$P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{1}{36}$$

Therefore, $P(\text{getting a doublet or a total of 4}) = P(A \cup B)$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= 6/36 + 3/36 - 1/36 = 8/36 = 2/9$$

Hence, the required probability is $2/9$.

Example

If A and B are two events such that $P(A) = 1/4$, $P(B) = 1/2$ and $P(A \text{ and } B) = 1/8$, find (i) $P(A \text{ or } B)$ (ii) $P(\text{not } A \text{ and not } B)$.

Solution

$$(i) P(A \text{ or } B) = P(A \cup B)$$

$$= P(A) + P(B) - P(A \cap B)$$

$$P(A \text{ or } B) = 1/4 + 1/2 - 1/8 = 5/8$$

$$(ii) P(\text{not } A \text{ and not } B) = P(\bar{A} \cap \bar{B})$$

$$= P(\overline{A \cup B})$$

$$= 1 - P(A \cup B)$$

$$P(\text{not } A \text{ and not } B) = 1 - 5/8 = 3/8$$

Example

A card is drawn from a pack of 52 cards. Find the probability of getting a king or a heart or a red card.

Solution

Total number of cards = 52; $n(S) = 52$

Suits of playing cards	Spade 	Heart 	Clavor 	Diamond 
Cards of each suit	A	A	A	A
	2	2	2	2
	3	3	3	3
	4	4	4	4
	5	5	5	5
	6	6	6	6
	7	7	7	7
	8	8	8	8
	9	9	9	9
	10	10	10	10
	J	J	J	J
	Q	Q	Q	Q
	K	K	K	K
Set of playing cards in each suit	13	13	13	13

Let A be the event of getting a king card. $n(A) = 4$

$$P(A) = \frac{n(A)}{n(S)} = \frac{4}{52}$$

Let B be the event of getting a heart card. $n(B) = 13$

$$P(B) = \frac{n(B)}{n(S)} = \frac{13}{52}$$

Let C be the event of getting a red card. $n(C) = 26$

$$P(C) = \frac{n(C)}{n(S)} = \frac{26}{52}$$

$$P(A \cap B) = P(\text{getting heart king}) = 1/52$$

$$P(B \cap C) = P(\text{getting red and heart}) = 13/52$$

$$P(A \cap C) = P(\text{getting red king}) = 2/52$$

$$P(A \cap B \cap C) = P(\text{getting heart, king which is red}) = 1/52$$

Therefore, required probability is

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

$$= 4/52 + 13/52 + 26/52 - 1/52 - 13/52 - 2/52 + 1/52 = 28/52$$

$$= \frac{4}{52} + \frac{13}{52} + \frac{26}{52} - \frac{1}{52} - \frac{13}{52} - \frac{2}{52} + \frac{1}{52} = \frac{28}{52} = \frac{7}{13}$$

$$= 7/13$$

Example:

The probability that a person will get an electrification contract is $3/5$ and the probability that he will not get plumbing contract is $5/8$. The probability of getting at least one contract is $5/7$. What is the probability that he will get both?

Solution:

Let A be the event of a person will get an electrification contract and B be a person will get plumbing contract.

$$P(A) = \frac{3}{5} \quad P(B') = \frac{5}{8} \quad \text{and} \quad P(A \cup B) = \frac{5}{7}$$

$$P(B) = 1 - P(B')$$

$$= 1 - \frac{5}{8} = \frac{3}{8}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\frac{5}{7} = \frac{3}{5} + \frac{3}{8} - P(A \cap B)$$

$$\frac{5}{7} = \frac{24 + 15}{40} - P(A \cap B)$$

$$\begin{aligned} P(A \cap B) &= \frac{39}{40} - \frac{5}{7} \\ &= \frac{273 - 200}{280} = \frac{73}{280} \end{aligned}$$

$$\text{Answer: } P(\text{getting both}) = \frac{73}{280}$$

Example. In a town of 8000 people, 1300 are over 50 years and 3000 are females. It is known that 30% of the females are over 50 years. What is the probability that a chosen individual from the town is either a female or over 50 years?

Solution:

Total People = 8000

$$n(S) = 8000$$

Let A be the event of choosing a females

$$n(A) = 3000$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{3000}{8000}$$

Let B be the event of choosing over 50 years

$$n(B) = 1300$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{1300}{8000}$$

$A \cap B$ = 30% females over 50 years

$$n(A \cap B) = \frac{30}{100} \times 3000 = 900$$

$$P(A \cap B) = \frac{900}{8000}$$

Using addition theorem of probability

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{3000}{8000} + \frac{1300}{8000} - \frac{900}{8000}$$

$$= \frac{3400}{8000} = \frac{34}{80} = \frac{17}{40}$$

Answer:

$$P(\text{either a female or over 50 years}) = \frac{17}{40}$$

Example

In a class of 50 students, 28 opted for NCC, 30 opted for NSS and 18 opted both NCC and NSS. One of the students is selected at random. Find the probability that

- (i) The student opted for NCC but not NSS.
- (ii) The student opted for NSS but not NCC.
- (iii) The student opted for exactly one of them.

Solution

Total number of students $n(S) = 50$.

Let A and B be the events of students opted for NCC and NSS respectively.

$$n(A) = 28, n(B) = 30, n(A \cap B) = 18$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{28}{50}$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{30}{50}$$

$$P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{18}{50}$$

- (i) Probability of the students opted for NCC but not NSS

$$P(A \cap \bar{B}) = P(A) - P(A \cap B) = 28/50 - 18/50 = 1/5$$

- (ii) Probability of the students opted for NSS but not NCC.

$$P(A \cap \bar{B}) = P(B) - P(A \cap B) = 30/50 - 18/50 = 6/25$$

- (iii) Probability of the students opted for exactly one of them

$$\begin{aligned}
 &= P[(A \cap \bar{B}) \cup (\bar{A} \cap B)] \\
 &= P(A \cap \bar{B}) + P(\bar{A} \cap B) = \frac{1}{5} + \frac{6}{25} = \frac{11}{25}
 \end{aligned}$$

(Note that $(A \cap \bar{B})$, $(\bar{A} \cap B)$ are mutually exclusive events)

Independent Events

Events are said to be independent if occurrence or non-occurrence of any one of the event does not affect the probability of occurrence or non-occurrence of the other events.

Two events A and B are said to be independent if and only if

$$P(A \cap B) = P(A) \cdot P(B)$$

Example

An unbiased die is thrown. If A is the event 'the number appearing is a multiple of 3' and B be the event 'the number appearing is even' number then find whether A and B are independent?

Solution:

We know that the sample space is $S = \{1,2,3,4,5,6\}$

Now, $A = \{3, 6\}$; $B = \{2,4,6\}$ then $(A \cap B) = \{6\}$

$$P(A) = \frac{2}{6} = \frac{1}{3}$$

$$P(B) = \frac{3}{6} = \frac{1}{2} \text{ and } P(A \cap B) = \frac{1}{6}$$

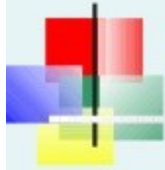
Clearly $P(A \cap B) = P(A) P(B)$

Hence A and B are independent events.

Formula for Joint Probability

Where:

- **$P(A \cap B)$** is the notation for the joint probability of event "A" and "B".
- **$P(A)$** is the probability of event "A" occurring.
- **$P(B)$** is the probability of event "B" occurring.



Joint Probability Example

P(Red and Ace)

$$= \frac{\text{number of cards that are red and ace}}{\text{total number of cards}} = \frac{2}{52}$$

Type	Color		Total
	Red	Black	
Ace	2	2	4
Non-Ace	24	24	48
Total	26	26	52

Event	Event		Total
	B ₁	B ₂	
A ₁	P(A ₁ and B ₁)	P(A ₁ and B ₂)	P(A ₁)
A ₂	P(A ₂ and B ₁)	P(A ₂ and B ₂)	P(A ₂)
Total	P(B ₁)	P(B ₂)	1

Joint Probabilities

Marginal (Simple) Probabilities

Conditional probability

The **conditional probability** of an event B , assuming that the event A has already happened is denoted by $P(B/A)$ and is defined as

$$P(B / A) = \frac{P(A \cap B)}{P(A)}, \text{ provided } P(A) \neq 0$$

Similarly,

$$P(A / B) = \frac{P(A \cap B)}{P(B)}, \text{ provided } P(B) \neq 0$$

(Multiplication theorem on probability)

The probability of the simultaneous happening of two events A and B is given by

$$\begin{array}{c} P(A \cap B) = P(A / B)P(B) \\ \text{or} \\ P(A \cap B) = P(B / A)P(A) \end{array}$$

Example

A number is selected randomly from the digits 11 through 19. Consider the events

$$A = \{ 11, 14, 16, 18, 19 \}$$

$$B = \{ 12, 14, 18, 19 \}$$

$$C = \{ 13, 15, 18, 19 \}.$$

Find (i) $P(A/B)$ (ii) $P(A/C)$ (iii) $P(B/C)$ (iv) $P(B/A)$

Solution:

$$\text{given } A = \{ 11, 14, 16, 18, 19 \} \quad B = \{ 12, 14, 18, 19 \} \quad C = \{ 13, 15, 18, 19 \}.$$

$$A \cap B = \{ 14, 18, 19 \} \quad A \cap C = \{ 18, 19 \} = B \cap C$$

$$P(A) = \frac{5}{9} \quad P(B) = \frac{4}{9} = P(C)$$

$$P(A \cap B) = \frac{3}{9} \quad P(A \cap C) = \frac{2}{9} = P(B \cap C)$$

Therefore, the probability for the occurrence of A given that B has occurred is

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{3}{9}}{\frac{4}{9}}$$

$$P(A/B) = \frac{3}{4}.$$

The probability for the occurrence of A given that C has occurred is

$$P(A/C) = \frac{P(A \cap C)}{P(C)} = \frac{\frac{2}{9}}{\frac{4}{9}}$$

$$P(A/C) = 1/2.$$

Similarly, the conditional probability of B given C is

$$P(B/C) = \frac{P(A \cap C)}{P(C)} = \frac{\frac{2}{9}}{\frac{4}{9}}$$

$$P(B/C) = \frac{1}{2}$$

and the conditional probability of B given A is

$$P(B/A) = \frac{P(B \cap A)}{P(A)} = \frac{\frac{3}{9}}{\frac{5}{9}}$$

$$P(B/A) = \frac{3}{5}.$$

Example A pair of dice is rolled and the faces are noted. Let

A: sum of the faces is odd, B: sum of the faces exceeds 8, and

C: the faces are different then find (i) $P(A/C)$ (ii) $P(B/C)$

Solution:

The outcomes favourable to the occurrence of these events are

$A = \{ (1,2), (1,4), (1,6), (2,1), (2,3), (2,5), (3,2), (3,4), (3,6), (4,1), (4,3), (4,5), (5,2), (5,4), (5,6), (6,1), (6,3), (6,5) \}$

$B = \{ (3,6), (4,5), (4,6), (5,4), (5,5), (5,6), (6,3), (6,4), (6,5), (6,6) \}$

$C = \{ (1,2), (1,3), (1,4), (1,5), (1,6), (2,1), (2,3), (2,4), (2,5), (2,6), (3,1), (3,2), (3,4), (3,5), (3,6), (4,1), (4,2), (4,3), (4,5), (4,6), (5,1), (5,2), (5,3), (5,4), (5,6), (6,1), (6,2), (6,3), (6,4), (6,5) \}$

Since A and B are proper subsets of C, $A \cap C = A$ and $B \cap C = B$.

$$\begin{aligned}\text{Here, } P(A) &= \frac{18}{36} = \frac{1}{2} \\ P(B) &= \frac{10}{36} = \frac{5}{9} \\ P(C) &= \frac{30}{36} = \frac{5}{6}.\end{aligned}$$

Since, $A \cap C = A$. Hence,

$$\begin{aligned}P(A \cap C) &= P(A) = \frac{1}{2} \\ P(B \cap C) &= P(B) = \frac{5}{9}.\end{aligned}$$

Hence, the probability for the sum of the faces is an odd number given that the faces are different is

$$P(A/C) = \frac{P(A \cap C)}{P(C)} = \frac{\frac{1}{2}}{\frac{5}{6}} = \frac{3}{5}$$

Similarly, the probability for the sum of the faces exceeds 8 given that the faces are different is

$$P(B/C) = \frac{P(B \cap C)}{P(C)} = \frac{\frac{5}{9}}{\frac{5}{6}} = \frac{4}{9}$$

Example

A bag contains 5 white and 3 black balls. Two balls are drawn at random one after the other without replacement. Find the probability that both balls drawn are black.

Solution:

Let A, B be the events of getting a black ball in the first and second draw
Probability of drawing a black ball in the first attempt is

$$P(A) = \frac{3}{5+3} = \frac{3}{8}$$

Probability of drawing the second black ball given that the first ball drawn is black

$$P(B/A) = \frac{2}{5+2} = \frac{2}{7}$$

$\therefore$ The probability that both balls drawn are black is given by

$$P(A \cap B) = P(A) P(B/A) = \frac{3}{8} \times \frac{2}{7} = \frac{3}{28}$$

Example

Find the probability of drawing a queen, a king and a knave (Jack) in that order from a pack of cards in three consecutive draws, the card drawn not being replaced.

Solution :

Let

A : the card drawn is a queen

B : the card drawn is a king

C : the card is drawn is a knave(jack)

$P(\text{drawing a queen card}) =$

$$P(\text{drawing a queen card}) = P(A) = \frac{4}{52}$$

$$P(\text{drawing a king card given that a queen card has been already drawn}) \\ = P(B/A) = \frac{4}{51}$$

$$P(\text{drawing a knave card given that a queen and a king cards have been drawn}) = P(C/AB) = \frac{4}{50}$$

Since they are dependent events, the required probability of the compound event is

$$P(ABC) = P(A) P(B/A) P(C/AB) = \frac{4}{52} \times \frac{4}{51} \times \frac{4}{50} = \frac{64}{132600} \\ = 0.00048$$

Example:

There are 4000 people living in a village including 1500 female. Among the people in the village, the age of 1000 people is above 25 years which includes 400 female. Suppose a person is chosen and you are told that the chosen person is a female. What is the probability that her age is above 25 years?

Solution:

Here, the event of interest is selecting a female with age above 25 years. In connection with the occurrence of this event, the following two events must happen.

A: a person selected is female

B: a person chosen is above 25 years.

Situation1:

We are interested in the event B , given that A has occurred. This event can be denoted by $B|A$. It can be read as “ B given A ”. It means that first the event A occurs then under that condition, B occurs. Here, we want to find the probability for the

occurrence of $B|A$ i.e., $P(B|A)$. This probability is called conditional probability. In reverse, the probability for selecting a female given that a person has been selected with age above 25 years is denoted by $P(A|B)$.

Situation 2:

Suppose that it is interested to select a person who is both female and with age above 25 years. This event can be denoted by $A \cap B$.

Calculation of probabilities in these situations warrant us to have another theorem namely Multiplication theorem. It is derived based on the definition of conditional probability.

$$P(A) = P(\text{Selecting a female}) = \frac{1500}{4000}$$

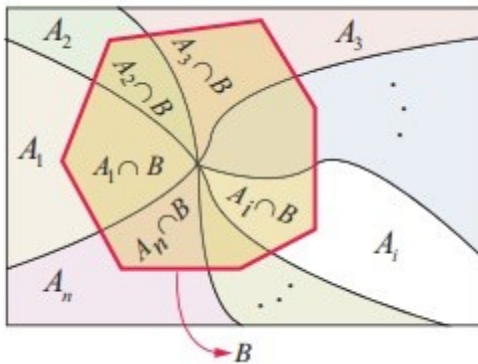
$$P(A \cap B) = P(\text{Selecting a female with age above 25 years}) = \frac{400}{1500}$$

Hence,

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{400}{1500} \times \frac{4000}{1500} = \frac{160}{225} = \frac{32}{45}.$$

Bayes' Theorem

Thomas Bayes was an English statistician, philosopher and Presbyterian minister who is known for formulating a specific case of a theorem. Bayesian methods stem from the principle of linking prior (before conducting experiment) probability and conditional probability (likelihood) to posterior (after conducting experiment) probability via Bayes' rule. Bayesian probability is the name given to several related interpretations of probability as an amount of epistemic confidence – the strength of beliefs, hypotheses etc., rather than a frequency.



1702-1761

Theorem (Bayes' Theorem)

If $A_1, A_2, A_3, \dots, A_n$ are mutually exclusive and exhaustive events such that $P(A_i) > 0, i = 1, 2, 3, \dots, n$ and B is any event in which $P(B) > 0$, then

$$P(A_i / B) = \frac{P(A_i) P(B / A_i)}{P(A_1) P(B / A_1) + P(A_2) P(B / A_2) + \dots + P(A_n) P(B / A_n)}$$

Proof

By the law of total probability of B we have

$P(B) = P(A_1) \cdot P(B / A_1) + P(A_2) \cdot P(B / A_2) + \dots + P(A_n) \cdot P(B / A_n)$ and by multiplication theorem $P(A_i \cap B) = P(B / A_i) P(A_i)$

By the definition of conditional probability,

$$P(A_i / B) = \frac{P(A_i \cap B)}{P(B)}$$

$$P(A_i / B) = \frac{P(B / A_i) P(A_i)}{P(A_1) \cdot P(B / A_1) + P(A_2) \cdot P(B / A_2) + \dots + P(A_n) \cdot P(B / A_n)}$$

(using formulae)

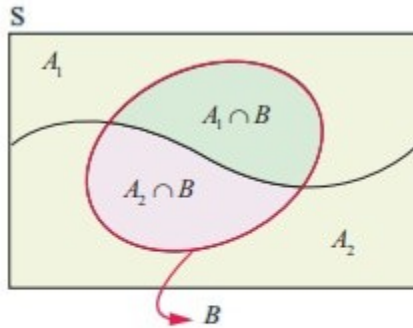
The above formula gives the relationship between $P(A_i / B)$ and $P(B / A_i)$

Example

A factory has two machines I and II. Machine I produces 40% of items of the output and Machine II produces 60% of the items. Further 4% of items produced by Machine I are defective and 5% produced by Machine II are defective. An item is drawn at random. If the drawn item is defective, find the probability that it was produced by Machine II. (See the previous example, compare the questions).

Solution

Let A_1 be the event that the items are produced by Machine-I, A_2 be the event that items are produced by Machine-II. Let B be the event of drawing a defective item. Now we are asked to find the conditional probability $P(A_2 / B)$. Since A_1, A_2 are mutually exclusive and exhaustive events, by Bayes' theorem,



$$P(A_2 / B) = \frac{P(A_2) P(B / A_2)}{P(A_1) P(B / A_1) + P(A_2) P(B / A_2)}$$

We have,

$$P(A_1) = 0.40, P(B / A_1) = 0.04$$

$$P(A_2) = 0.60, P(B / A_2) = 0.05$$

$$P(A_2 / B) = \frac{P(A_2) P(B / A_2)}{P(A_1) P(B / A_1) + P(A_2) P(B / A_2)}$$

$$P(A_2 / B) = \frac{(0.60)(0.05)}{(0.40)(0.04) + (0.60)(0.05)} = \frac{15}{23}.$$

Example

A construction company employs 2 executive engineers. Engineer-1 does the work for 60% of jobs of the company. Engineer-2 does the work for 40% of jobs of the company. It is known from the past experience that the probability of an error when engineer-1 does the work is 0.03, whereas the probability of an error in the work of engineer-2 is 0.04. Suppose a serious error occurs in the work, which engineer would you guess did the work?

Solution

Let A_1 and A_2 be the events of job done by engineer-1 and engineer-2 of the company respectively. Let B be the event that the error occurs in the work.

We have to find the conditional probability

$P(A_1 / B)$ and $P(A_2 / B)$ to compare their errors in their work.

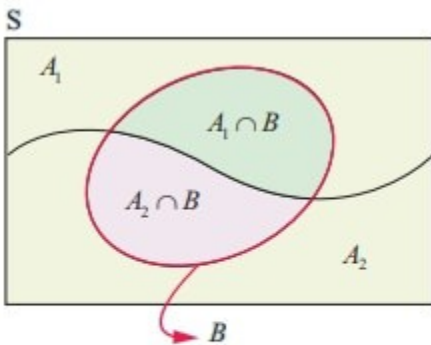
From the given information, we have

$$P(A_1) = 0.60, P(B / A_1) = 0.03$$

$$P(A_2) = 0.40, P(B / A_2) = 0.04$$

A_1 and A_2 are mutually exclusive and exhaustive events.

Applying Bayes' theorem,



$$P(A_1 / B) = \frac{P(A_1) P(B / A_1)}{P(A_1) P(B / A_1) + P(A_2) P(B / A_2)}$$
$$= \frac{(0.60)(0.03)}{(0.60)(0.03) + (0.40)(0.04)}$$

$$P(A_1 / B) = \frac{9}{17}.$$

$$P(A_2 / B) = \frac{P(A_2) P(B / A_2)}{P(A_1) P(B / A_1) + P(A_2) P(B / A_2)}$$

$$P(A_2 / B) = \frac{(0.40)(0.04)}{(0.60)(0.03) + (0.40)(0.04)}$$

$$P(A_2 / B) = \frac{8}{17}.$$

Since $P(A_1 / B) > P(A_2 / B)$, the chance of error done by engineer-1 is greater than the chance of error done by engineer-2. Therefore one may guess that the serious error would have been done by engineer-1.

Example

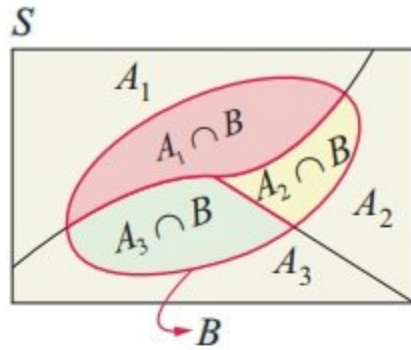
The chances of X , Y and Z becoming managers of a certain company are $4 : 2 : 3$. The probabilities that bonus scheme will be introduced if X , Y and Z become managers are 0.3 , 0.5 and 0.4 respectively. If the bonus scheme has been introduced, what is the probability that Z was appointed as the manager?

Solution

Let A_1 , A_2 and A_3 be the events of X , Y and Z becoming managers of the company respectively. Let B be the event that the bonus scheme will be introduced.

We have to find the conditional probability $P (A_3 / B)$.

Since A_1 , A_2 and A_3 are mutually exclusive and exhaustive events, applying Bayes' theorem



We have

$$P(A_3 / B) = \frac{P(A_3) P(B / A_3)}{P(A_1) P(B / A_1) + P(A_2) P(B / A_2) + P(A_3) P(B / A_3)}$$

$$P(A_1) = \frac{4}{9}, \quad P(B / A_1) = 0.3$$

$$P(A_2) = \frac{2}{9}, \quad P(B / A_2) = 0.5$$

$$P(A_3) = \frac{3}{9}, \quad P(B / A_3) = 0.4$$

$$P(A_3 / B) = \frac{P(A_3) P(B / A_3)}{P(A_1) P(B / A_1) + P(A_2) P(B / A_2) + P(A_3) P(B / A_3)}$$

$$\begin{aligned} P(A_3 / B) &= \frac{\left(\frac{3}{9}\right)(0.4)}{\left(\frac{4}{9}\right)(0.3) + \left(\frac{2}{9}\right)(0.5) + \frac{3}{9}(0.4)} \\ &= \frac{12}{34} = \frac{6}{17}. \end{aligned}$$

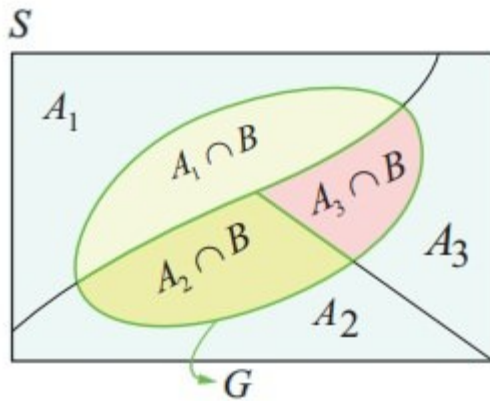
Example

A consulting firm rents car from three agencies such that 50% from agency L , 30% from agency M and 20% from agency N . If 90% of the cars from L , 70% of cars from M and 60% of the cars from N are in good conditions (i) what is the

probability that the firm will get a car in good condition? (ii) if a car is in good condition, what is probability that it has come from agency N ?

Solution

Let A_1 , A_2 , and A_3 be the events that the cars are rented from the agencies X , Y and Z respectively.



Let G be the event of getting a car in good condition.

We have to find

(i) the total probability of event G that is, $P(G)$

(ii) find the conditional probability A_3 given G that is, $P(A_3 / G)$ We have

$$P(A_1) = 0.50, P(G / A_1) = 0.90$$

$$P(A_2) = 0.30, P(G / A_2) = 0.70$$

$$P(A_3) = 0.20, P(G / A_3) = 0.60.$$

(i) Since A_1 , A_2 and A_3 are mutually exclusive and exhaustive events and G is an event in S , then the total probability of event G is $P(G)$.

$$P(G) = P(A_1) P(G / A_1) + P(A_2) P(G / A_2) + P(A_3) P(G / A_3)$$

$$P(G) = (0.50)(0.90) + (0.30)(0.70) + (0.20)(0.60)$$

$$P(G) = 0.78.$$

(ii) The conditional probability A_3 given G is $P(A_3 / G)$

By Bayes' theorem,

$$P(A_3 / G) = \frac{P(A_3) P(G / A_3)}{P(A_1) P(G / A_1) + P(A_2) P(G / A_2) + P(A_3) P(G / A_3)}$$

$$P(A_3 / G) = \frac{(0.20)(0.60)}{(0.50)(0.90) + (0.30)(0.70) + (0.20)(0.60)}$$

$$= \frac{2}{13}.$$

EXERCISE

(1) A factory has two Machines-I and II. Machine-I produces 60% of items and Machine-II produces 40% of the items of the total output. Further 2% of the items produced by Machine-I are defective whereas 4% produced by Machine-II are defective. If an item is drawn at random what is the probability that it is defective?

(2) There are two identical urns containing respectively 6 black and 4 red balls, 2 black and 2 red balls. An urn is chosen at random and a ball is drawn from it. (i) find the probability that the ball is black (ii) if the ball is black, what is the probability that it is from the first urn?

(3) A firm manufactures PVC pipes in three plants viz, X, Y and Z. The daily production volumes from the three firms X, Y and Z are respectively 2000 units, 3000 units and 5000 units. It is known from the past experience that 3% of the output from plant X, 4% from plant Y and 2% from plant Z are defective. A pipe is selected at random from a day's total production,

(i) find the probability that the selected pipe is a defective one.

(ii) if the selected pipe is a defective, then what is the probability that it was produced by plant Y?

(4) The chances of A , B and C becoming manager of a certain company are $5 : 3 : 2$. The probabilities that the office canteen will be improved if A , B , and C become managers are 0.4 , 0.5 and 0.3 respectively. If the office canteen has been improved, what is the probability that B was appointed as the manager?

(5) An advertising executive is studying television viewing habits of married men and women during prime time hours. Based on the past viewing records he has determined that during prime time wives are watching television 60% of the time. It has also been determined that when the wife is watching television, 40% of the time the husband is also watching. When the wife is not watching the television, 30% of the time the husband is watching the television. Find the probability that (i) the husband is watching the television during the prime time of television (ii) if the husband is watching the television, the wife is also watching the television.

Events	Explanation	Example
Equally likely events	Two or more events are said to be equally likely if each one of them has an equal chance of occurring.	Head and tail are equally likely events in tossing a coin .
Certain events	In an experiment, the event which surely occur is called certain event .	When we roll a die , the event of getting any natural number from one to six is a certain event.
Impossible events	In an experiment if an event has no scope to occur then it is called an impossible event .	When we toss two coins , the event of getting three heads is an impossible event.
Mutually exclusive events	Two or more events are said to be mutually exclusive if they don't have common sample points. i.e., events A , B are said to be mutually exclusive if $A \cap B = \phi$.	When we roll a die the events of getting odd numbers and even numbers are mutually exclusive events.
Exhaustive events	The collection of events whose union is the whole sample space are called exhaustive events .	When we toss a coin twice , the collection of events of getting two heads, exactly one head, no head are exhaustive events.
Complementary events	The complement of an event A is the event representing collection of sample points not in A. It is denoted A' or A^c or $\bar{A}$ The event A and its complement A' are mutually exclusive and exhaustive.	When we roll a die , the event 'rolling a 5 or 6' and the event of rolling a 1, 2, 3 or 4 are complementary events.

EXERCISE 12.1

1. An experiment has the four possible mutually exclusive and exhaustive outcomes A, B, C, and D. Check whether the following assignments of probability are permissible.

(i) $P(A) = 0.15,$ $P(B) = 0.30,$
 $P(C) = 0.43,$ $P(D) = 0.12$

(ii) $P(A) = 0.22,$ $P(B) = 0.38,$
 $P(C) = 0.16,$ $P(D) = 0.34$

(iii) $P(A) = \frac{2}{5},$ $P(B) = \frac{3}{5},$
 $P(C) = -\frac{1}{5},$ $P(D) = \frac{1}{5}$

Solution :

(i) $P(A) = 0.15,$ $P(B) = 0.30,$ $P(C) = 0.43,$
 $P(D) = 0.12$

Given $P(A) = 0.15, P(B) = 0.30, P(C) = 0.43,$
 $P(D) = 0.12$

$P(A), P(B), P(C) \text{ and } P(D) \geq 0$

Also, $P(S) = P(A) + P(B) + P(C) + P(D)$
 $= 0.15 + 0.30 + 0.43 + 0.12 = 1$

$\therefore$ The assignment of probability is permissible.

(ii) $P(A) = 0.22,$ $P(B) = 0.38,$ $P(C) = 0.16,$
 $P(D) = 0.34$

Given $P(A) = 0.22 \geq 0, P(B) = 0.38 \geq 0,$

$P(C) = 0.16 \geq 0$ and $P(D) = 0.34 \geq 0$

Also, $P(S) = 0.22 + 0.38 + 0.16 + 0.34 = 1.1 \neq 1$

$\therefore$ The assignment of probability is not permissible.

(iii) $P(A) = \frac{2}{5},$ $P(B) = \frac{3}{5},$ $P(C) = -\frac{1}{5},$
 $P(D) = \frac{1}{5}$

$$\Rightarrow n(S) = 4$$

Let A be the event of getting one head and one tail

$$\therefore A = \{HT, TH\}$$

$$\Rightarrow n(A) = 2$$

$$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{2}{4} = \frac{1}{2}$$

(ii) Let B be the event of getting atmost two tails

$$\therefore B = \{HH, HT, TH, TT\}$$

$$\therefore n(B) = 4$$

$$\therefore P(B) = \frac{n(B)}{n(S)} = \frac{4}{4} = 1$$

- 3.** Five mangoes and 4 apples are in a box. If two fruits are chosen at random, find the probability that (i) one is a mango and the other is an apple (ii) both are of the same variety.

Solution : $S = \{ 5 \text{ mangoes and } 4 \text{ apples} \}$

$$\Rightarrow n(S) = {}^9C_2$$

[$\because$ 2 fruits are taken from 9 fruits]

(i) Let A be the event of getting one mango and one apple

$$\therefore n(A) = {}^5C_1 \times {}^4C_1 = 5 \times 4 = 20$$

$$\therefore P(A) = \frac{20}{{}^9C_2} = \frac{20}{\frac{9 \times 8}{2 \times 1}} = \frac{20 \times 2}{9 \times 8} = \frac{5}{9}$$

(ii) Let B be the event of getting 2 fruits from the same variety.

$$\therefore n(B) = {}^5C_2 + {}^4C_2$$

$$= \frac{5 \times 4}{2 \times 1} + \frac{4 \times 3}{2 \times 1} = \frac{10}{2} + \frac{6}{2}$$

1 2 3 4 5
 Sunday, Monday, Tuesday, Wednesday, Thursday

6 7
 Friday or Saturday

∴ Probability that an ordinary year may contain 53

$$\text{Sundays} = \frac{1}{7}$$

(ii) A leap year contains 366 days.

$$366 \text{ days} = 52 \text{ Weeks} + 2 \text{ days.}$$

These two days can be selected from the following combinations.

- 1) Monday & Tuesday
- 2) Tuesday & Wednesday
- 3) Wednesday & Thursday
- 4) Thursday & Friday
- 5) Friday & Saturday
- 6) Saturday & Sunday
- 7) Sunday & Monday

$$\therefore \text{Required probability} = \frac{\text{No. of favourable events}}{\text{Total no. of events}} = \frac{2}{7}$$

5. Eight coins are tossed once, find the probability of getting

- (i) exactly two tails (ii) at least two tails
 (iii) at most two tails

Solution : Since 8 coins are tossed,

$$(i) \quad n(S) = 2^8 = 256$$

Let A be the event of getting exactly two tails

$$\Rightarrow \therefore n(A) = {}^8C_2 = \frac{{}^4P_2 \times 7}{2 \times 1} = 4 \times 7$$

(iii) Let C be the event of getting at most two tails

$$\begin{aligned}\therefore n(C) &= {}^8C_0 + {}^8C_1 + {}^8C_2 = 1 + 8 + \frac{8 \times 7}{2 \times 1} \\ &= 9 + 28 = 37 \\ \therefore P(C) &= \frac{n(C)}{n(S)} = \frac{37}{256}\end{aligned}$$

6. An integer is chosen at random from the first 100 positive integers. What is the probability that the integer chosen is a prime or multiple of 8?

Solution : Let $S = \{1, 2, \dots, 100\}$

$$\Rightarrow n(S) = 100$$

Let A be the event of getting a prime number and B be the event of getting multiple of 8.

$$\therefore A = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97\}$$

$$\Rightarrow n(A) = 25$$

$$B = \{8, 16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96\}$$

$$\Rightarrow n(B) = 12$$

$$\therefore n(A \text{ or } B) = n(A) + n(B) = 25 + 12 = 37$$

$$\therefore P(A \text{ or } B) = \frac{n(A) + n(B)}{n(S)} = \frac{37}{100}$$

[$\because$ A and B are mutually exclusive events, $A \cap B = \phi$]

7. A bag contains 7 red and 4 black balls, 3 balls are drawn at random. Find the probability that (i) all are red (ii) one red and 2 black.

Solution : $S = \{7 \text{ Red, } 4 \text{ Black balls}\}$

$$\Rightarrow n(S) = {}^{11}C_3$$

[$\because$ 3 balls are drawn out of 11 balls]

(i) Let A be the events of getting 3 red balls

$$\therefore P(A) = \frac{n(B)}{n(S)} = \frac{42}{{}^{11}C_3} = \frac{42}{\frac{11 \times 10 \times 9}{3 \times 2 \times 1}} = \frac{42 \times 2}{11 \times 10} = \frac{14}{55}$$

8. A single card is drawn from a pack of 52 cards. What is the probability that

- (i) the card is an ace or a king
- (ii) the card will be 6 or smaller
- (iii) the card is either a queen or 9?

Solution : $S = \{\text{Pack of 52 cards}\}$

$$\therefore n(S) = 52$$

(i) $P(\text{ace card or a king card}) = P(\text{ace card}) + P(\text{king card})$
[Since they are mutually exclusive]

$$= \frac{4}{52} + \frac{4}{52} = \frac{8}{52} = \frac{2}{13}$$

(ii) $P(\text{card will be 6 or smaller})$

$$= \frac{5 + 5 + 5 + 5}{52} = \frac{20}{52} = \frac{5}{13}$$

[$\because$ 5 cards which are 6 or smaller from each variety]

$$(iii) P(\text{queen card or 9}) = \frac{4}{52} + \frac{4}{52} = \frac{8}{52} = \frac{2}{13}$$

9. A cricket club has 16 members, of whom only 5 can bowl. What is the probability that in a team of 11 members at least 3 bowlers are selected?

Solution : $S = \{\text{Members of Cricket Club}\}$

$$\therefore n(S) = {}^{16}C_{11}$$

[$\because$ 11 members must be selected for a team]

Let A be the event of selecting at least 3 bowlers.

At least 3 bowlers can be selected as follows :

$$= \frac{16 \times 15 \times 14 \times 13 \times 12}{8 \times 4 \times 8 \times 2} = 4368$$

$$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{3762}{4368} = \frac{627}{728}$$

10. (i) The odds that the event A occurs is 5 to 7, find P(A).

(ii) Suppose $P(B) = \frac{2}{5}$. Express the odds that the event B occurs.

Solution : The odds of an event A are $a : b$ in favour of an event and

$$P(A) = \frac{a}{a+b}$$

(i) Given that odds of an event A occurs is 5 : 7

$$\therefore P(A) = \frac{5}{5+7} = \frac{5}{12}$$

(ii) Given $P(B) = \frac{2}{5} = \frac{2}{2+3}$ [$\because a = 2$ and $b = 3$]

$\therefore$ The odds that the event B occurs is 2 to 3.

Probability Distribution

A **probability distribution** is a mathematical function that, stated in simple terms, can be thought of as providing the probabilities of occurrence of different possible outcomes in an experiment. For instance, if the random variable X is used to denote the outcome of a coin toss ("the experiment"), then the probability distribution of X would take the value 0.5 for X = heads, and 0.5 for X = tails (assuming the coin is fair).

A probability distribution is a statistical function that describes all the possible values and likelihoods that a random variable can take within a given range. This range will be between the minimum and maximum statistically possible values, but where the possible value is likely to be plotted on the probability distribution depends on a number of factors. These factors include the distribution's mean, standard deviation, skewness and kurtosis.

An example will make clear the relationship between random variables and probability distributions. Suppose you flip a coin two times. This simple statistical experiment can have four possible outcomes: HH, HT, TH, and TT. Now, let the variable X represent the number of Heads that result from this experiment. The variable X can take on the values 0, 1, or 2. In this example, X is a random variable; because its value is determined by the outcome of a statistical experiment.

Types of Probability Distribution

There are two types of probability distributions:

1. Discrete probability distributions
2. Continuous probability distributions

Discrete random variable

A random variable is said to be discrete if it takes only a finite or countable infinite number of values.

Example

Consider the experiment of tossing a coin

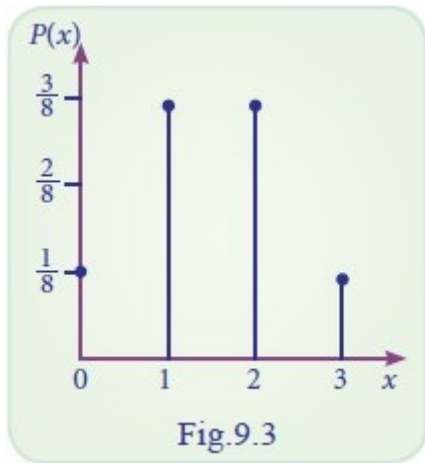
If $X(\text{Head}) = 1$, $X(\text{Tail}) = 0$

Then X takes the values either 0 or 1

This is a discrete random variable.

Probability mass function and probability density function

A probability function is associated with each value of the random variable. This function is used to compute probabilities for events associated with the random variables. The probability function defined for a discrete random variable is called probability mass function. The probability function associated with continuous random variable is called probability density function.



Probability Mass Function

If, X is a discrete random variable taking values $x_1, x_2, \dots, x_n$ with respective probabilities $p(x_1), p(x_2), \dots, p(x_n)$ such that

(i) $p(x_i) \geq 0, \forall i$ (non-negative)

and (ii) $\sum_{i=1}^n p(x_i) = 1$

then $p(x)$ is known as the probability mass function (p.m.f) of the discrete random variable X .

The pair $\{x_i, p(x_i); i = 1, 2, 3, \dots\}$ is known as probability distribution of X .

Example

A coin is tossed two times. If X is the number of heads, find the probability mass function of X .

Solution:

Since the coin is tossed two times, the sample space is $S = \{HH, HT, TH, TT\}$

If X denotes the numbers of heads, the possible values of X are 0,1,2 with the following

$$P(X = 0) = P(\text{getting no head}) = \frac{1}{4}$$

$$P(X = 1) = P(\text{getting one head}) = \frac{2}{4} = \frac{1}{2}$$

$$P(X = 2) = P(\text{getting two heads}) = \frac{1}{4}$$

The probability distribution of X is

X	0	1	2
$p(X = x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

Example

In example 9.3 the probability mass function of X is given in the following table

X	2	3	4	5	6	7	8	9	10	11	12
$P(x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

The above table may be called as the probability distribution function of X .

Continuous random variable

Definition

A random variable X which can take on any value (integral as well as fraction) in the interval is called continuous random variable.

Examples of continuous random variable

The amount of water in a 10 ounce bottle.

The speed of a car.

Electricity consumption in kilowatt hours.

Height of people in a population.

Weight of students in a class.

The length of time taken by a truck driver to go from Chennai to Madurai, etc.

Probability density function

Definition

The probability that a random variable X takes a value in the interval $[t_1, t_2]$ (open or closed) is given by the integral of a function called the probability density function $f_x(x)$:

$$P(t_1 \leq X \leq t_2) = \int_{t_1}^{t_2} f_x(x) dx .$$

$$P(t_1 \leq X \leq t_2) = \int_{t_1}^{t_2} f_X(x) dx .$$

Other names that are used instead of probability density function include density function, continuous probability function, integrating density function.

The probability density functions $f_x(x)$ or simply by $f(x)$ must satisfy the following conditions.

(i) $f(x) \geq 0 \forall x$ and

(ii) $\int_{-\infty}^{\infty} f(x) dx = 1 .$

\\

Example

A continuous random variable X has the following p.d.f

$$f(x) = ax, 0 \leq x \leq 1$$

Determine the constant a and also find $P [X \leq 1/2]$

Solution:

We know that

$$\begin{aligned}\int_{-\infty}^{\infty} f(x)dx &= 1 \\ \int_0^1 ax \, dx &\Rightarrow a \int_0^1 x dx = 1 \\ &\Rightarrow a \left(\frac{x^2}{2} \right)_0^1 = 1 \\ &\Rightarrow \frac{a}{2}(1-0) = 1 \\ &\Rightarrow a = 2\end{aligned}$$

$$\begin{aligned}P\left[x \leq \frac{1}{2}\right] &= \int_{-\infty}^{\frac{1}{2}} f(x)dx \\ &= \int_0^{\frac{1}{2}} ax dx \\ &= \int_0^{\frac{1}{2}} 2x dx \\ &= \frac{1}{4}\end{aligned}$$

Example

A continuous random variable X has p.d.f

$$f(x) = 5x^4, 0 \leq x \leq 1$$

Find a_1 and a_2 such that (i) $P[X \leq a_1] = P[X > a_1]$ (ii) $P[X > a_2] = 0.05$

Solution

(i) Since $P[X \leq a_1] = P[X > a_1]$

$$\begin{aligned}
 P[X \leq a_1] &= \frac{1}{2} \\
 \text{i.e., } \int_0^{a_1} f(x) dx &= \frac{1}{2} \\
 \text{i.e., } \int_0^{a_1} 5x^4 dx &= \frac{1}{2} \\
 5 \left[\frac{x^5}{5} \right]_0^{a_1} &= \frac{1}{2} \\
 a_1 &= (0.5)^{\frac{1}{5}}
 \end{aligned}$$

(ii) $P[X > a_2] = 0.05$

$$\begin{aligned}
 \int_{a_2}^1 f(x) dx &= 0.05 \\
 \int_{a_2}^1 5x^4 dx &= 0.05 \\
 5 \left[\frac{x^5}{5} \right]_{a_2}^1 &= 0.05 \\
 a_2 &= [0.95]^{\frac{1}{5}}
 \end{aligned}$$

Continuous distribution function

Definition

If X is a continuous random variable with the probability density function $f_x(x)$, then the function $F_x(x)$ is defined by

$F_x(x) = P[X \leq x] = \int_{-\infty}^x f(t) dt$, $-\infty < x < \infty$ is called the distribution function (d.f) or sometimes the cumulative distribution function (c.d.f) of the continuous random variable X

Properties of cumulative distribution function

The function $F_x(x)$ or simply $F(x)$ has the following properties

(i) $0 \leq F(x) \leq 1, -\infty < x < \infty$

(ii) $F(-\infty) = \lim_{x \rightarrow -\infty} F(x) = 0$ and $F(+\infty) = \lim_{x \rightarrow \infty} F(x) = 1.$

(iii) $F(\cdot)$ is a monotone, non-decreasing function; that is, $F(a) \leq F(b)$ for $a < b$.

(iv) $F(\cdot)$ is continuous from the right; that is, $\lim_{h \rightarrow 0} F(x+h) = F(x).$

(v) $F'(x) = \frac{d}{dx} F(x) = f(x) \geq 0$

(vi) $F'(x) = \frac{d}{dx} F(x) = f(x) \Rightarrow dF(x) = f(x)dx$

$dF(x)$ is known as probability differential of X .

(vii)
$$\begin{aligned} P(a \leq x \leq b) &= \int_a^b f(x)dx = \int_{-\infty}^b f(x)dx - \int_{-\infty}^a f(x)dx \\ &= P(X \leq b) - P(X \leq a) \\ &= F(b) - F(a) \end{aligned}$$

$$= P(X \leq b) - P(X \leq a)$$

$$= F(b) - F(a)$$

Example

Suppose, the life in hours of a radio tube has the following p.d.f

$$f(x) = \begin{cases} \frac{100}{x^2}, & \text{when } x \geq 100 \\ 0, & \text{when } x < 100 \end{cases}$$

Find the distribution function.

Solution:

$$\begin{aligned} F(x) &= \int_{-\infty}^x f(t) dt \\ &= \int_{100}^x \frac{100}{t^2} dt, x \geq 100 \\ &= \left[\frac{100}{-t} \right]_{100}^x, x \geq 100 \\ F(x) &= \left[1 - \frac{100}{x} \right], x \geq 100 \end{aligned}$$

Example

The amount of bread (in hundreds of pounds) x that a certain bakery is able to sell in a day is found to be a numerical valued random phenomenon, with a probability function specified by the probability density function $f(x)$ is given by

$$f(x) = \begin{cases} Ax, & \text{for } 0 \leq x < 10 \\ A(20 - x), & \text{for } 10 \leq x < 20 \\ 0, & \text{otherwise} \end{cases}$$

- (a) Find the value of A .
- (b) What is the probability that the number of pounds of bread that will be sold tomorrow is
 - (i) More than 10 pounds,
 - (ii) Less than 10 pounds, and

(iii) Between 5 and 15 pounds?

Solution:

(a) We know that

$$\int_{-\infty}^{\infty} f(x)dx = 1$$
$$\int_0^{10} Axdx + \int_{10}^{20} A(20-x)dx = 1$$
$$A \left\{ \left[\frac{x^2}{2} \right]_0^{10} + \left[20x - \frac{x^2}{2} \right]_{10}^{20} \right\} = 1$$

$$A[(50 - 0) + (400 - 200) - (200 - 50)] = 1$$

$$A = \frac{1}{100}$$

(b) (i) The probability that the number of pounds of bread that will be sold tomorrow is more than 10 pounds is given by

$$\begin{aligned} P(10 \leq X \leq 20) &= \int_{10}^{20} \frac{1}{100} (20-x) dx \\ &= \frac{1}{100} \left[20x - \frac{x^2}{2} \right]_{10}^{20} \\ &= \frac{1}{100} [(400 - 200) - (200 - 50)] \\ &= 0.5 \end{aligned}$$

(ii) The probability that the number of pounds of bread that will be sold tomorrow is less than 10 pounds, is given by

$$\begin{aligned}
 P(0 \leq X < 10) &= \int_0^{10} \frac{1}{100} x dx \\
 &= \frac{1}{100} \left[\frac{x^2}{2} \right]_0^{10} \\
 &= \frac{1}{100} (50 - 0) \\
 &= 0.5
 \end{aligned}$$

(iii) The probability that the number of pounds of bread that will be sold tomorrow is between 5 and 15 pounds is

$$\begin{aligned}
 P(5 \leq X \leq 15) &= \int_5^{10} \frac{1}{100} x dx + \int_{10}^{15} \frac{1}{100} (20 - x) dx \\
 &= \frac{1}{100} \left[\frac{x^2}{2} \right]_5^{10} + \frac{1}{100} \left[20x - \frac{x^2}{2} \right]_{10}^{15} \\
 &= 0.75
 \end{aligned}$$

Joint and marginal probability mass functions

In real life situations we may be observed two or more random variables on the individuals simultaneously. For instance, blood pressure and cholesterol for each individual are measured simultaneously. In such cases we require the concept of bi-variate random variable represented by (X, Y) , where X and Y are univariate random variables.

Definition (Joint p.m.f)

Let (X, Y) be a discrete bivariate random variable. Then $p(x, y)$ is called the joint probability mass function of (X, Y) if the following conditions are satisfied.

$$p(x, y) \geq 0 \quad \forall x, y$$

$$\sum_{x,y} p(x,y) = 1$$

Definition (Marginal Probability Mass Function)

Given a joint probability mass function $p(x, y)$, then $p(x) = \sum_y p(x,y)$ is called marginal probability mass function of X . Similarly $p(y) = \sum_x p(x,y)$ is called the marginal probability mass function of Y .

Example

There are 10 tickets in a bag which are numbered 1, 2, 3, ...10. Two tickets are drawn at random one after the other with replacement.

Here, random variable X denotes the number on the first ticket and random variable Y denotes the number on the second ticket.

Joint and marginal probability density functions

As we defined in section 9.5.1 the joint probability mass function, we define the joint probability density function.

Definition:

Let (X, Y) be a bivariate continuous random variables. The function $f(x, y)$ is called a bivariate probability density if the following conditions are satisfied.

$$(i) \quad f(x, y) \geq 0 \quad \forall x, y$$

$$(ii) \quad \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

The marginal probability density function of X is given by

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

and the marginal probability density function of Y is given by

$$h(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

Example

Prove that the bivariate function given by $f(x, y) = \begin{cases} \frac{1}{8}(x+y) & , \quad 0 < x, y \leq 2 \\ 0 & , \quad \text{otherwise} \end{cases}$

Proof:

If f is a probability density function

$$\begin{aligned} & \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1 \\ &= \frac{1}{8} \int_0^2 \int_0^2 (x+y) dx dy \\ &= \frac{1}{8} \left(\int_0^2 \int_0^2 x dx dy + \int_0^2 \int_0^2 y dx dy \right) \\ &= \frac{1}{8} \left\{ \int_0^2 \left[\int_0^2 x dx \right] dy + \int_0^2 \left[\int_0^2 y dy \right] dx \right\} \\ &= \frac{1}{8} \left[\int_0^2 2 dy + \int_0^2 2 dx \right] \\ &= \frac{1}{8} [(2y)_0^2 + (2x)_0^2] \\ &= \frac{1}{8} [(4-0) + (4-0)] \\ &= \frac{1}{8} \times 8 = 1 \end{aligned}$$

Therefore, $f(x, y)$ is a probability density function.

Example 9.16

Joint p.d.f. of X, Y is $f(x, y) = \begin{cases} x^2 y & , \quad 0 \leq x \leq 1, 0 \leq y \leq 2 \\ 0 & , \quad \text{elsewhere} \end{cases}$

then find the marginal density function of X and Y .

Solution:

$$\begin{aligned}
 f(x) &= \int_{-\infty}^{\infty} f(x, y) dy \\
 f(x) &= \int_1^2 x^2 y dy \\
 &= x^2 \left[\frac{y^2}{2} \right]_1^2 = x^2 \left[\frac{4}{2} - \frac{1}{2} \right] \\
 &= x^2 \times \frac{3}{2} = \frac{3}{2} x^2 \\
 f(y) &= \int_{-\infty}^{\infty} f(x, y) dx \\
 f(y) &= \int_0^1 x^2 y dy \\
 &= y \left(\frac{x^3}{3} \right)_0^1 = y \left(\frac{1}{3} \right) = \frac{y}{3}
 \end{aligned}$$

Marginal density function of X

$$f(x) = \begin{cases} \frac{3}{2}x^2 & , \quad 0 \leq x \leq 1, \\ 0 & , \quad elsewhere \end{cases}$$

Marginal density function of Y

$$f(y) = \begin{cases} \frac{y}{3} & , \quad 0 \leq y \leq 2 \\ 0 & , \quad elsewhere \end{cases}$$

Example

Joint p.d.f. of X, Y is $f(x, y) = \begin{cases} 4x - 2y & , \quad 0 \leq x \leq 3, 0 \leq y \leq 2 \\ 0 & , \quad elsewhere \end{cases}$
density function of X and Y

Find the marginal

Solution:

$$f(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

$$\begin{aligned} f(x) &= \int_1^2 (4x - 2y) dy \\ &= \left[4xy - \frac{2y^2}{2} \right]_1^2 \\ &= [4x(2 - 1) - (4 - 1)] = 4x - 3 \end{aligned}$$

$$f(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

$$\begin{aligned} f(y) &= \int_1^3 (4x - 2y) dx \\ &= \left(\frac{4x^2}{2} - 2xy \right)_1^3 \\ &= (2x^2 - 2xy)_1^3 \\ &= [2(9 - 1) - 2y(3 - 1)] \\ &= 2 \times 8 - 6y + 2y \\ &= 16 - 4y \end{aligned}$$

Marginal density function of X

$$f(x) = \begin{cases} 4x - 3 & , \quad 1 \leq x \leq 3 \\ 0 & , \quad elsewhere \end{cases}$$

Marginal density function of Y

$$f(x, y) = \begin{cases} 16 - 4y & , \quad 1 \leq y \leq 2 \\ 0 & , \quad elsewhere \end{cases}$$

